

Solutions End Sem M3 (25 Nov 2024)

1a) $f(x) = x \log_{10} x - 1.2$; $f'(x) = \log_{10} x + \log_{10} e$

$$f(2) < 0 \quad \& \quad f(3) > 0$$

So by Intermediate Value Theorem, root lies b/w 2 & 3.

Taking $x_0 = 2$ & using Newton Raphson method

$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}$$

we get $x_1 = 2.81316$

$$x_2 = 2.74110$$

$$x_3 = 2.74064$$

$$x_4 = 2.74064$$

Ans 2.7406

1b) Rearranging to make the system diagonally dominant

$$5x - y + z = 10$$

$$2x + 4y = 12$$

$$x + y + 5z = -1$$

$$\left. \begin{array}{l} 5x - y + z = 10 \\ 2x + 4y = 12 \\ x + y + 5z = -1 \end{array} \right\} \Rightarrow \begin{array}{l} x = \frac{1}{5}(10 + y - z) \\ y = \frac{1}{2}(6 - x) \\ z = \frac{1}{5}(-1 - x - y) \end{array}$$

Taking $(x^{(0)}, y^{(0)}, z^{(0)}) = (0.5, 0.5, 0.5)$ using Jacobi method we have

$$x^{(n+1)} = \frac{1}{5}(10 + y^{(n)} - z^{(n)})$$

$$y^{(n+1)} = \frac{1}{2}(6 - x^{(n)})$$

$$z^{(n+1)} = \frac{1}{5}(-1 - x^{(n)} - y^{(n)})$$

which gives

	$x^{(n)}$	$y^{(n)}$	$z^{(n)}$
n=1	2.00000	2.75000	-0.40000
n=2	2.63000	2.00000	-1.15000
n=3	2.63000	1.68500	-1.12600
n=4	2.56220	1.68500	-1.06300
n=5	2.54960	1.71890	-1.04944
n=6	2.55366	1.72520	-1.05370
n=7	2.55578	1.72316	-1.05577
n=8	2.55578	1.72211	-1.05578
n=9	2.55558	1.72210	-1.05558
n=10	2.55553	1.72221	-1.05553

Ans $x = 2.5555$

$$y = 1.7222$$

$$z = -1.0555$$

2a)

x	y	Δy	$\Delta^2 y$	$\Delta^3 y$	$\Delta^4 y$
-4	1245				
		-404			
-1	33		94		
		-28		-14	
0	5		10		3
		2		13	
2	9		88		
		442			
5	1335				

Using Newton's divided difference formula

$$y = y_0 + (x-x_0) \Delta y_0 + (x-x_0)(x-x_1) \Delta^2 y_0 + \dots + (x-x_0)(x-x_1)(x-x_2)(x-x_3) \Delta^4 y_0$$

gives

$$y = 3x^4 - 5x^3 + 6x^2 - 14x + 5$$

Thus $y(6) = 2945 //$

2b) Given

	x_0	x_1	x_2	x_3
<u>x</u>	20	25	30	35
<u>y</u>	0.342	0.423	0.500	0.650
	y_0	y_1	y_2	y_3

Using Lagrange's Inverse Interpolation Formula we have

$$x = \frac{(y-y_1)(y-y_2)(y-y_3)}{(y_0-y_1)(y_0-y_2)(y_0-y_3)} x_0 + \frac{(y-y_0)(y-y_2)(y-y_3)}{(y_1-y_0)(y_1-y_2)(y_1-y_3)} x_1 + \frac{(y-y_0)(y-y_1)(y-y_3)}{(y_2-y_0)(y_2-y_1)(y_2-y_3)} x_2 + \frac{(y-y_0)(y-y_1)(y-y_2)}{(y_3-y_0)(y_3-y_1)(y_3-y_2)} x_3$$

which gives for $y = 0.390$, $x = 23.67 //$ Answer

3a)

	0	135				
			14			
	300	149	-6			
			8	24		
	600	157	18	-50		
			26	-26	70	
x_0	900	183	-8	20	-86	
			18	-6	-16	
	1200	201	-14	4		
			4	-2		
	1500	205	-16			
			-12			
	1800	193				

Difference Table

Stirling's Formula

$$\begin{aligned}
 y = & y_0 + p \left(\frac{\Delta y_{-1} + \Delta y_0}{2} \right) + \frac{p^2}{2!} \Delta^2 y_{-1} + \frac{p(p^2-1)}{3!} \left(\frac{\Delta^3 y_{-2} + \Delta^3 y_{-1}}{2} \right) \\
 & + \frac{p^2(p^2-1)}{4!} \Delta^4 y_{-2} + \frac{p(p^2-1)(p^2-4)}{5!} \left(\frac{\Delta^5 y_{-3} + \Delta^5 y_{-2}}{2} \right) \\
 & + \frac{p^2(p^2-1)(p^2-4)}{6!} \Delta^6 y_{-3}
 \end{aligned}$$

where $p = \frac{x - x_0}{h}$

$$\begin{aligned}
 \Rightarrow \frac{dy}{dx} = & \frac{1}{h} \left(\frac{\Delta y_{-1} + \Delta y_0}{2} + p \Delta^2 y_{-1} + \frac{(3p^2-1)}{6} \left(\frac{\Delta^3 y_{-2} + \Delta^3 y_{-1}}{2} \right) \right. \\
 & + \frac{(2p^3-p)}{12} \Delta^4 y_{-2} + \frac{(5p^4-15p^2+4)}{120} \left(\frac{\Delta^5 y_{-3} + \Delta^5 y_{-2}}{2} \right) \\
 & \left. + \frac{(3p^5-10p^3+4p)}{360} \Delta^6 y_{-3} \right)
 \end{aligned}$$

$$\& \frac{d^2 y}{dx^2} = \frac{1}{h^2} \left(\Delta^2 y_{-1} + p \left(\frac{\Delta^3 y_{-2} + \Delta^3 y_{-1}}{2} \right) + \frac{6p^2 - 1}{12} \Delta^4 y_{-2} \right. \\ \left. + \frac{2p^3 - 3p}{12} \left(\frac{\Delta^5 y_{-3} + \Delta^5 y_{-2}}{2} \right) + \frac{15p^4 - 20p + 4}{360} \Delta^6 y_{-3} \right)$$

At $x = 900$ we have $p = 0$

Hence

$$\left. \frac{dy}{dx} \right|_{x=900} = \frac{1}{h} \left(\frac{\Delta y_{-1} + \Delta y_0}{2} - \frac{1}{6} \left(\frac{\Delta^3 y_{-2} + \Delta^3 y_{-1}}{2} \right) \right. \\ \left. + \frac{1}{30} \left(\frac{\Delta^5 y_{-3} + \Delta^5 y_{-2}}{2} \right) \right)$$

$$\& \left. \frac{d^2 y}{dx^2} \right|_{x=900} = \frac{1}{h^2} \left(\Delta^2 y_{-1} - \frac{1}{12} \Delta^4 y_{-2} + \frac{1}{90} \Delta^6 y_{-3} \right)$$

Substituting values we have

$$\text{Velocity at } 900s = \frac{1}{300} \left(\frac{26+18}{2} - \frac{(-26-6)}{12} + \frac{(70-16)}{60} \right) \\ = 0.0852 \text{ m/s} //$$

$$\text{Acceleration at } 900s = \frac{1}{300^2} \left(-8 - \frac{20}{12} - \frac{86}{90} \right) \\ = -0.0001 \text{ m/s}^2 //$$

3b) 6 subintervals $\Rightarrow n=6, x_0=0, x_6=6$
 $\Rightarrow h = \frac{6-0}{6} = 1$

x	0	1	2	3	4	5	6
$y = \frac{1}{1+x^2}$	1	$\frac{1}{2}$	$\frac{1}{5}$	$\frac{1}{10}$	$\frac{1}{17}$	$\frac{1}{26}$	$\frac{1}{37}$

1) By Trapezoidal Rule

$$\int_0^6 \frac{dx}{1+x^2} = \frac{h}{2} (y_0 + y_6 + 2(y_1 + \dots + y_5))$$

$$= 1.4108$$

2) By Simpson's $\frac{1}{3}$ Rule

$$\int_0^6 \frac{dx}{1+x^2} = \frac{h}{3} (y_0 + y_6 + 4(y_1 + y_3 + y_5) + 2(y_2 + y_4))$$

$$= 1.3662$$

Actual answer $\int_0^6 \frac{dx}{1+x^2} = \tan^{-1} 6 = 1.4056$

Error by Trapezoidal Rule = 0.0052

Error by Simpson's $\frac{1}{3}$ Rule = 0.0394

Hence in this case, Trapezoidal Rule is giving a better answer.

Though in general the magnitude of error in Simpson's $\frac{1}{3}$ is of order h^2 compared to order h for Trapezoidal Rule.

4a) Given $y' = -xy^2$, $y(2)=1$ & $h=0.1$

$$x_0=2, y_0=1, f(x,y)=-xy^2 \Rightarrow x_1=2.1, x_2=2.2$$

By Euler's method

$$y_1 = y_0 + h f(x_0, y_0) \quad y_1^{(0)} = \mathbf{0.80000}$$

Using modified Euler's method

$$y_1^{(n+1)} = y_0 + \frac{h}{2} (f(x_0, y_0) + f(x_1, y_1^{(n)}))$$

$$y_1^{(1)} = \mathbf{0.83280} \quad y_1^{(2)} = \mathbf{0.82717} \quad y_1^{(3)} = \mathbf{0.82815}$$

$$y_1^{(4)} = \mathbf{0.82798} \quad y_1^{(5)} = \mathbf{0.82801} \quad y_1^{(6)} = \mathbf{0.82801}$$

$$\text{Thus } y_1 = 0.8280$$

By Euler's method

$$y_2 = y_1 + h f(x_1, y_1)$$

Using modified Euler's method $y_2^{(0)} = \mathbf{0.68403}$

$$y_2^{(n+1)} = y_1 + \frac{h}{2} (f(x_1, y_1) + f(x_2, y_2^{(n)}))$$

$$y_2^{(1)} = \mathbf{0.70455} \quad y_2^{(2)} = \mathbf{0.70141} \quad y_2^{(3)} = \mathbf{0.70190}$$

$$y_2^{(4)} = \mathbf{0.70182} \quad y_2^{(5)} = \mathbf{0.70184} \quad y_2^{(6)} = \mathbf{0.70183}$$

$$\text{Thus } y(2.2) = \mathbf{0.7018} \quad \underline{\underline{\text{Answer}}}$$

4b) Given $h=0.2$ & $y' = \frac{2xy + e^x}{x^2 + xe^x}$, $x(1)=0$

$$x_0=1, y_0=0, x_1=1.2, f(x,y) = \frac{2xy + e^x}{x^2 + xe^x}$$

Using R-K method of fourth order

$$k_1 = h f(x_0, y_0) \quad k_1 = \mathbf{0.14621}$$

$$k_2 = h f\left(x_0 + \frac{h}{2}, y_0 + \frac{k_1}{2}\right) \quad k_2 = \mathbf{0.14021}$$

$$k_3 = h f\left(x_0 + \frac{h}{2}, y_0 + \frac{k_2}{2}\right) \quad k_3 = \mathbf{0.13991}$$

$$k_4 = h f(x_0 + h, y_0 + k_3) \quad k_4 = \mathbf{0.13480}$$

$$y_1 = y_0 + \frac{1}{6}(k_1 + 2k_2 + 2k_3 + k_4) \quad y_1 = \mathbf{0.14021}$$

$$x_2 = x_1 + h = 1.4$$

Using R-K method of fourth order

$$k_1 = h f(x_1, y_1) \quad k_1 = 0.13482$$

$$k_2 = h f\left(x_1 + \frac{h}{2}, y_1 + \frac{k_1}{2}\right) \quad k_2 = 0.13031$$

$$k_3 = h f\left(x_1 + \frac{h}{2}, y_1 + \frac{k_2}{2}\right) \quad k_3 = 0.13013$$

$$k_4 = h f(x_1 + h, y_1 + k_3) \quad k_4 = 0.12601$$

$$y_2 = y_1 + \frac{1}{6}(k_1 + 2k_2 + 2k_3 + k_4) \quad y_2 = 0.27050$$

Thus $y(1.4) = 0.2705$ // Answer

5a)

x	y	$x - \bar{x}$	$y - \bar{y}$	$(x - \bar{x})(y - \bar{y})$	$(x - \bar{x})^2$	$(y - \bar{y})^2$
70	25.2	-5	-3.14	15.7	25	9.8596
75	28.6	0	0.26	0	0	0.0676
80	30.2	5	1.86	9.3	25	3.4596
60	22.3	-15	-6.04	90.6	225	36.4816
90	35.4	15	7.06	105.9	225	49.8436
$\bar{x} = 75$	$\bar{y} = 28.34$			221.5	500	99.7120

$$\text{Thus } r = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum (x_i - \bar{x})^2} \sqrt{\sum (y_i - \bar{y})^2}} = 0.992$$

As the correlation coefficient is almost 1, the relation between x and y is almost linear and y is directly proportional to x .

Line of regression of y on x

$$(y - \bar{y}) = b_{xy} (x - \bar{x})$$

$$\text{where } b_{xy} = r \frac{\sqrt{\sum (y_i - \bar{y})^2}}{\sqrt{\sum (x_i - \bar{x})^2}} = 0.443$$

Hence line becomes

$$(y - 28.34) = 0.443(x - 75)$$

i.e. $y = 0.443x - 4.885 //$

Given $x = 100$

we have $y = 39.415 //$

5b) To fit $y = ax^2 + bx + c$ using least squares we minimise

$$E = \sum (y - y_i)^2 = \sum (ax_i^2 + bx_i + c - y_i)^2$$

The normal eqⁿs are

$$\frac{\partial E}{\partial a} = 0, \quad \frac{\partial E}{\partial b} = 0, \quad \frac{\partial E}{\partial c} = 0$$

$$\Rightarrow a \sum x_i^4 + b \sum x_i^3 + c \sum x_i^2 - \sum x_i^2 y_i = 0$$

$$a \sum x_i^3 + b \sum x_i^2 + c \sum x_i - \sum x_i y_i = 0$$

$$a \sum x_i^2 + b \sum x_i + nc - \sum y_i = 0$$

(n=7)

x	y	x^2	x^3	x^4	xy	x^2y
1.0	1.1	1.00	1.000	1.0000	1.10	1.100
1.5	1.3	2.25	3.375	5.0625	1.95	2.925
2.0	1.6	4.00	8.000	16.0000	3.20	6.400
2.5	2.0	6.25	15.625	39.0625	5.00	12.500
3.0	2.7	9.00	27.000	81.0000	8.10	24.300
3.5	3.4	12.25	42.875	150.0625	11.90	41.650
4.0	4.1	16.00	64.000	256.0000	16.40	65.600
Sum	17.5	16.2	50.75	161.875	47.65	154.475

$$\Rightarrow 548.1875a + 161.875b + 50.75c = 154.475$$

$$161.875a + 50.75b + 17.5c = 47.65$$

$$50.75a + 17.5b + 7c = 16.2$$

Solving we get

$$a = 0.242857$$

$$b = -0.192857$$

$$c = 1.035714$$

Thus the required curve is

$$y = 0.242857x^2 - 0.192857x + 1.035714 //$$